

Mukesh Kumar

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Summary

Lead Software Engineer with 11+ years of experience building highly scalable & distributed systems across fintech, logistics, and e-commerce. Currently operating at Staff-level, leading org-wide initiatives in schema governance, Infrastructure-as-a-Service (IaaS), Identity & Access Management (IAM) systems etc. Strong focus on scalability, distributed systems, operational excellence, and cross-functional technical leadership. Exploring applied AI to build intelligent platforms by combining LLMs with scalable distributed systems.

Professional Experience

10/2022 – Present
Bangalore India

Phonepe, Lead Software Engineer | Platform as a Service (PaaS) Team

- **Schemate (Schema governance):** Defined and led org-wide schema governance architecture for event-driven systems across 50+ microservices and event processing workloads.
 - Built schema evolution framework with backward/forward compatibility validation, schema rollback, and governance policies for event-driven systems.
 - Fast tracked adoption across **50+ microservices** and stream processing at **3M RPS** to filter schema-compatible events; reduced schema-related production incidents **60%** in 9 months.
- **Garuda (IaaS):** Architected and drove adoption of an internal IaaS platform enabling infrastructure governance and standardized operational workflows across engineering teams.
 - Reducing SRE toil by **80% (~200 hrs/week)** and improving infra reliability to 99.99% for mission-critical DB operations, security upgrades, DR activities, and infrastructure management.
- **OlympusIM (Identity Platform):** Extended org-wide AuthN, AuthZ, and RBAC to **10K RPS** with **99.99% SLO**.
 - Led tracks for multi-tenant roles, cascading auth identity for service-to-service and async communications, and Maker-Checker for privileged access; Drove Java 17 upgrade and production hardening.
- **AI Analytics Copilot:** Built an AI-powered analytics assistant pet project for Business and product teams, enabling natural language query of event-driven data stored in ElasticSearch. Implemented semantic schema discovery using metadata-based RAG, automatic query generation, conversational analytics, and AI-driven insights with interactive visualizations.
 - Designed a scalable architecture separating metadata retrieval from events store, enabling efficient analytics over large-scale datasets without embedding raw events.

07/2020 – 09/2022
Bangalore, India

Udaan, Senior Software Engineer (L3) | Udaan Express B2B Logistics

- Increased Udaan-Express daily orders from **1K to 200K** across **1K clients** within 6 months by optimizing API integrations and bulk-order processing throughput.
- Improved platform stability from **99.9% to 99.99%** while cost-effectively sustaining **2K RPS** through performance monitoring and reliability tuning.
- Implemented variable rate-card pricing for lane-based and zone-based B2B/B2C clients, enabling accurate and scalable logistics charge calculations.
- Enabled cashless COD payments through a verified payment-link system sent to consignees, improving payment success and customer checkout experience.
- Built a push-based notification platform on Azure Event Hub to deliver real-time order updates to sellers.
- Automated the DPM incentive system with contract pricing based on KM, KG, TP, and FUEL metrics, eliminating manual field executive calculations.

09/2019 – 07/2020
Gurugram, India

OYO, *Senior Software Engineer | Payment Gateways | OYO Money Wallet*

- Designed wallet redemption with user-segment rules, increasing customer bookings **by 3–5%**.
- Built on-page OTP collection for Paytm gateway hotel payments, improving payment success **by 2–3%**.

08/2015 – 06/2019
Gurugram, India

Delhivery, *Senior Software Developer | Logistics services Client Panel*

- Led full-stack development of Client Panel enabling 10K+ clients to manifest, track orders, and generate customized reports.
- Implemented bulk order processing to increase 100K/daily order to 200K+/daily across 10+clients.
- Implemented multi-user management for the client with role-based access control for core platform services.
- Built an actionable insights dashboard with WoW/MoM analytics across ETA/TAT, order volume, revenue, and geographic shipment mapping.
- Delivered a comprehensive reporting system with scheduled report generation and NDR shipment tracking.

Software Developer

- Built **Hyperlocal**, a web-based ERP solution for inventory & order management, enabling local retailers across Delhivery's 10K+ client ecosystem.
- Core capabilities - real-time inventory visibility, automated replenishment triggers, and integrated supply chain. Reducing manual errors and improving fulfillment predictability.

01/2015 – 06/2015

Elucidata (formerly Jeevomics) , *Software Developer Intern*
Life Sciences Platform

- Built an interactive Experiments and Analysis Tracker for **General Metabolics** scientists, improving visibility into experiment lifecycle, research data, and analysis progress for cross-functional science teams.

Skills

- **Core Expertise:** Distributed systems, Event-driven systems, Microservices, Platform strategy, cross-functional technical leadership.
- **Languages & frameworks:** Java (primary), Kotlin, Python, JavaScript, Dropwizard, Spring Boot, Nodejs
- **Data & infra:** MariaDB, PostgreSQL, NoSQL DB, DynamoDB, Redis, Aerospike, Kafka, Kubernetes, Docker, AWS, Azure, CI/CD
- **Identity & reliability:** IAM Systems, JWT, RBAC, Multi-tenancy, Maker-Checker, Grafana, Opsgenie.
- **AI & GenAI:** LLM Applications, Retrieval-Augmented Generation (RAG), LangGraph, Vector Search, Prompt Engineering

Education

B.Tech, Information Technology — IIIT Allahabad · 2011–2015 · GPA 8.49 / 10

10+2 CBSE — JNV · Shahjahanpur, India · 2009 - 2010 · GPA: 90.8%

Awards

- **Merit Incentive Award** — IIIT Allahabad · Recognized for consistent academic excellence (2011–2015)
- **CBSE Computer Science Certificate of Merit** — Top **0.1%** nationwide · 99% in Computer Science (10+2)
- **CBSE Board Merit Scholarship** — Awarded Senior Secondary merit scholarship for outstanding board performance